

Magnifying Facial Micro-movements for Cognitive Evaluation

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Abstract—Objective cognitive assessment is critical for the early detection and management of cognitive decline. The Mini-Mental State Examination (MMSE) is a widely used tool for this purpose, but it requires face-to-face interaction and manual scoring by clinicians. Recent advances in computer vision and deep learning offer the potential to automate and enhance the accuracy of such assessments. This study presents a novel deep learning model that integrates multimodal data captured during cognitive testing sessions on a tablet. By focusing on facial movements, which are captured and magnified through a pre-processing pipeline, the model classifies inputs into categories corresponding to MMSE scores. Our results show a significant correlation between facial movements and MMSE, suggesting the feasibility of using automated video analysis as a reliable proxy for cognitive assessment.

Index Terms—MMSE, facial micro-movements, motion magnification, deep learning

I. INTRODUCTION

Facial expression is one of the primary communication channels for humans. Recent successes in encoding facial muscle patterns, such as Facial Action Units (AUs) [1], and their use in studying attention [2], [3], affect [4], pain [5], and language-related pathologies [6] demonstrate a correlation between facial movements/expressions and cognitive states. Recently, several studies have also analyzed patterns of emotional expression in subjects with mild cognitive impairment (MCI) or severe cognitive impairment associated with Alzheimer’s disease (AD). For instance, in [7], the authors investigated facial expression imitation abilities in a sample of subjects with MCI and AD using data from a 3D Kinect. No statistically significant differences were observed between healthy individuals and MCI patients, while specific (and statistically significant) imitation patterns were identified in the AD group.

However, within the field of cognitive neuroscience, it has been highlighted that ordinary facial expressions, i.e., macro-

expressions, do not always reflect a person’s true emotions. Conversely, the emergence of involuntary micro-expressions is considered a significant clue for revealing cognitive states related to the suppression of subjective feelings [8] or neurocognitive disorders [9], [10]. Micro-expressions are brief, subtle, and involuntary facial expressions. Unlike macro-expressions, micro-expressions, being involuntary and non-conscious movements, last only 40 to 200 ms [11] and are too rapid to be captured by a human observer or even a standard video camera.

These considerations provide motivation to explore the dynamics of facial muscle movements in various neurocognitive disorders and correlate them with the patients’ cognitive state (Mini-Mental State Examination, MMSE [12]), in the context of remote monitoring of the cognitive state of subjects with MCI. However, commercial video cameras, as well as devices like the 3D Kinect used in [7], do not have a temporal resolution sufficient to detect facial micro-movements, and the use of higher-resolution cameras is not feasible due to increasing costs. To overcome these limitations, we propose to extract micro-movements from low-resolution videos, appropriately modified and enhanced using Phase-Based Video Magnification (PBVM) [13], a method proposed to reveal subtle changes in the movement of objects in videos that are impossible to see with the naked eye.

To summarize, in this paper we propose a deep learning-based approach for estimating MMSE scores, using videos of subjects performing cognitive tasks and serious games [14] on a tablet device for rehabilitation purposes. Our model isolates and enhances facial micro-movements and combines them with additional inputs, including emotional states inferred from video recordings and task performance metrics. Our research aims to establish a correlation between facial micro-movements and MMSE scores, supporting the hypothesis that automated video analysis can serve as a non-invasive and scalable proxy for cognitive assessment.

II. METHOD

A. Architecture Overview

The architecture of the proposed approach consists of two main stages, as illustrated in Fig. 1: a *visual pre-processing*

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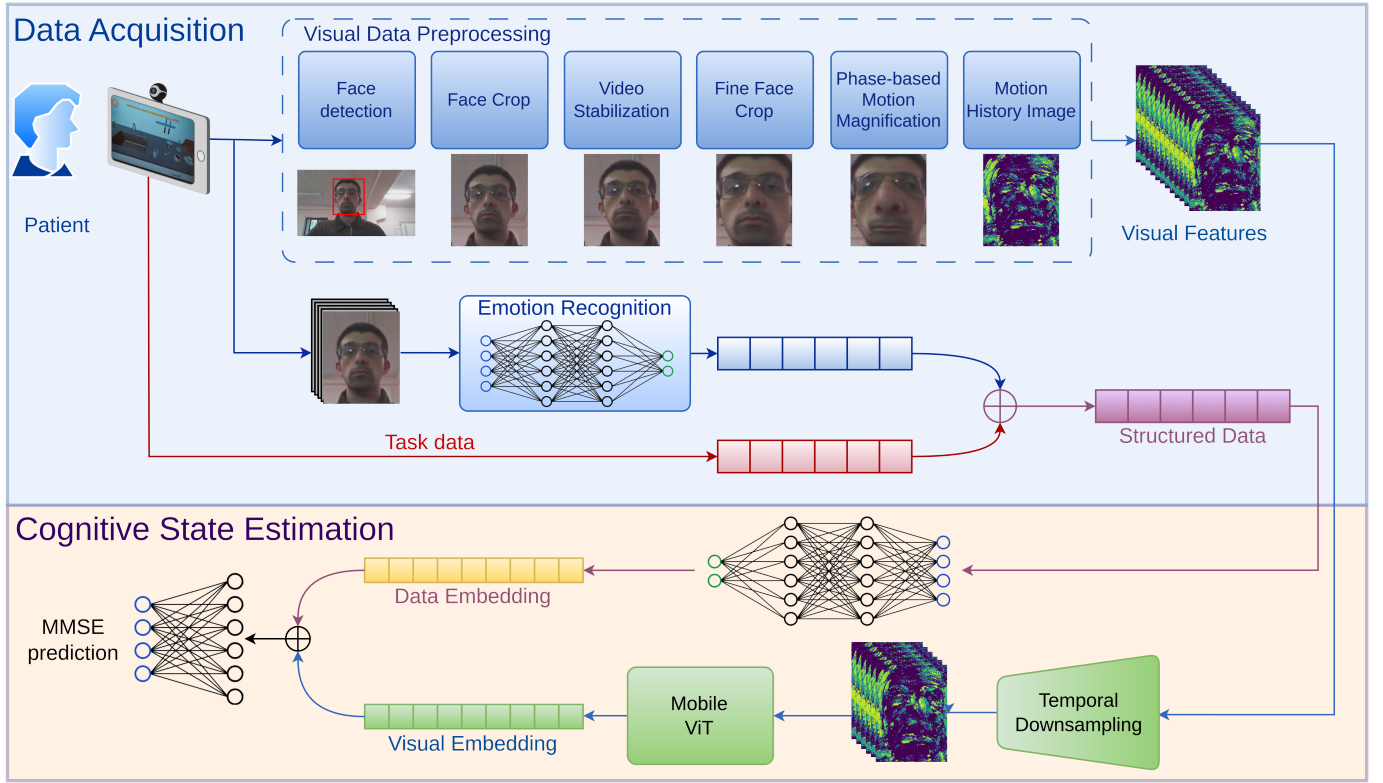


Fig. 1. Overview of the proposed approach. Facial micro-expression features are extracted through a visual pre-processing pipeline and combined with emotional states and performance metrics on cognitive tasks to predict a subject’s MMSE score.

pipeline, which analyzes facial features from a video to capture and emphasize facial micro-movements, and a *cognitive state estimation* deep network, which combines facial micro-expressions with emotional states and performance metrics on cognitive tasks to predict the subject’s MMSE score.

B. Visual Pre-processing

The pre-processing of video data is a critical step in our study, as it ensures that the input to our deep learning model is of high quality and suitable for the analysis of facial micro-movements and expressions. Our pre-processing pipeline receives a video clip of a subject performing a cognitive task and consists of several stages:

1. **Face detection:** We apply a Multi-task Cascaded Convolutional Network (MTCNN) [15] to detect faces within the video frames. MTCNN is a three-stage deep learning model that effectively identifies facial structures with high accuracy, even under different lighting and pose conditions.
2. **Main face selection:** If multiple subjects are detected, we use a heuristic based on proximity to the tablet and the size of the detected face to select the primary subject, presumed to be the patient.
3. **Emotion recognition:** We adopt HyperExtended Light-Face [16] to perform frame-by-frame emotion recognition, classifying each detected face into one of seven emotional states (anger, disgust, fear, happiness, sadness,

surprise, neutral). This stage produces a 7D vector, where each component encodes the prevalence of each emotion as the ratio of video frames in which that emotion is recognized.

4. **Face crop:** In order to maintain a consistent focus on the face region throughout the analysis and remove unnecessary information, we crop the subject’s face in each frame. The crop region extends the face bounding box by 25% vertically and horizontally, to allow for slight movements.
5. **Video stabilization:** We employ the Good Features to Track (GFTT) detector [17] from the VidStab library to mitigate the effects of involuntary patient motion and stabilize the background. This technique corrects for frame-to-frame jitter and provides a stable foundation for accurate motion magnification.
6. **Fine face crop:** Post-stabilization, we perform a second round of face detection using MTCNN to refine the region of interest, cropping the face bounding box. This step ensures that the focus remains tightly on the facial area after any adjustments made during stabilization.
7. **Motion magnification:** We apply a Phase-Based Motion Magnification algorithm [18] to amplify subtle facial movements that are otherwise imperceptible. The algorithm decomposes video frames into a complex steerable pyramid and computes the local phase over time at every spatial scale and orientation. These phases are tempo-

rally bandpassed to isolate specific temporal frequencies, corresponding to motion in different spatial scales and orientations. Magnified motion is then computed by amplifying the bandpassed phases by a factor α . The entire procedure can be summarized as follows

$$\Phi_{\text{amplified}} = \Phi + \alpha \cdot (\Phi - \Phi_{\text{filtered}}), \quad (1)$$

where Φ is the original phase of the image pyramid coefficients, Φ_{filtered} is the temporally filtered phase, and α is the amplification factor. The amplified phase $\Phi_{\text{amplified}}$ is then used with the original amplitude to reconstruct the motion-magnified video signal.

8. **Motion History Image:** To capture the temporal aspect of facial movements, we generate a Motion History Image (MHI) [19] for each 3-second window of the video. MHI is a visual representation of motion where more recent movements are brighter, and older movements fade over time: it is updated at each time step t by overlaying the binary motion image $D(x, y, t)$ onto the MHI from the previous time step $H_{t-1}(x, y)$. The updating rule for the MHI is as follows:

$$H_t(x, y) = \begin{cases} \tau & \text{if } D(x, y, t) = 1, \\ \max(0, H_{t-1}(x, y) - \delta) & \text{otherwise.} \end{cases} \quad (2)$$

where τ is a constant intensity value assigned at (x, y) locations where motion is detected (i.e., where $D(x, y, t) = 1$) and δ is the decay rate, which controls how quickly the intensity value decreases over time in the absence of new motion. Using MHI allows for efficiently aggregating motion information into a compact representation, increasing the temporal context for MMSE prediction. Examples of MHI outputs are reported in Fig. 2.

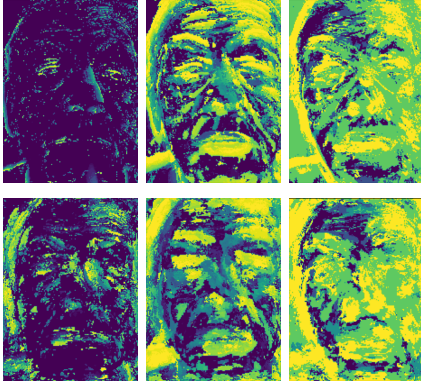


Fig. 2. Examples of motion history images at different temporal granularities. The three columns represent the effects using windows of 1, 10 and 60 seconds. The top row shows the videos after face stabilization and cropping, and the bottom row shows the same videos after motion magnification.

C. Cognitive State Estimation

In order to effectively exploit the rich data collected while a subject carries out cognitive tasks on a tablet device, the proposed cognitive state estimation model features a two-branch processing architecture. The first branch is designed

to manage the structured data that comes directly from the interactive tasks, including standardized indices for sessions and runs (indicating the degree of familiarity of the subject with the exercises and the number of exercises performed within the session, reflecting tiredness), performance scores from the cognitive exercises, an identifier for each specific exercise, and the duration of the exercises. This information is also integrated by emotion prevalence vectors, described above. We concatenate these inputs into a 12-dimensional vector and feed it to a fully-connected network, generating a 256-dimensional embedding that encapsulates structured task data.

The visual branch processes variable-length video frames through a trainable *temporal downsampling module*, which receives the sequence of motion history images and feeds them to a cascade of two 3D convolutions operating on the temporal dimension alone. As output this module provides an image-like tensor of resolution $H \times W \times T$, with H , W and T made independent of the original input size by means of adaptive max pooling. These features are then fed to a pre-trained Mobile ViT [20]; the class token is used as an aggregate representation and projected to a 256-dimensional visual embedding.

Both 256-dimensional embeddings from structured data and visual features are concatenated to form a 512-dimensional feature vector, which is then processed through two fully-connected layers to predict the subject's MMSE score.

III. EXPERIMENTAL RESULTS

A. Dataset

The dataset was collected from a clinical trial involving 23 patients undergoing rehabilitation sessions. The study sample consisted of 13 male and 10 female participants from two different hospitals, providing a variety of clinical representations.

The number of sessions per patient ranged from 5 to 13, each lasting approximately 40 minutes and taking place on different days. This variability reflects the individual differences in cognitive decline and rehabilitation progress. Each rehabilitation session consisted of 8 to 17 exercises, indicating the patients' level of participation in the rehabilitation program. The frequency and duration of the sessions were adjusted based on the patients' cognitive abilities and speed. Exercises were selected from a set of seven distinct cognitive tasks, categorized into four serious games related to activities of daily living and three additional cognitive tasks, providing a variety of stimuli and cognitive challenges that are critical for evaluating the rehabilitation progress of patients.

Data collection involved recording videos of the patients during each session. This provided a dual modality for a comprehensive analysis of patient interaction and cognitive performance. A total of 162 sessions was recorded, resulting in 2294 individual exercise runs, each lasting between 1 and 10 minutes.

The patient sample showed mild cognitive impairment, with MMSE scores ranging from 20 to 28. We also recorded the score obtained in each of question of the MMSE described

in [12]. The dataset’s range of MMSE scores enables a nuanced analysis of cognitive responses in correlation with varying degrees of cognitive function.

B. Training and Evaluation

We split the dataset into training, validation and test sets with proportion 56/22/22, such that all sessions of the same subject belong to a single split.

We train the proposed model using the AdamW [21] optimizer with a weight decay of $5 \cdot 10^{-4}$, a learning rate of $5 \cdot 10^{-5}$, and a batch size of 12. Parameters α , τ and δ are set, respectively, to 600, 255, 5. The temporal downsampling module provides an output of 16 frames at 224×224 resolution. Following the literature on the prediction of non-categorical ordered values [22], we minimize a weighted cross-entropy loss, accounting for the class imbalance within the dataset. Class labels correspond to MMSE scores of patients in our dataset (20 to 28). Training was conducted until convergence, and model selection is performed based on validation accuracy.

Results on MMSE estimation are reported in Tab. I, in terms of balanced accuracy, sensitivity, specificity and F1 score. Employing task data and emotions alone already provides above-chance prediction accuracy: the former encodes information on the capability of a subject to solve a task, which affects the duration of the exercise; the latter reflects the serenity or frustration of the subject, which correlates with the relative task complexity. Visual features alone, however, outperform the usage of task data and emotions, demonstrating the effectiveness of facial expressions in assessing cognitive capabilities. Importantly, the model’s accuracy significantly benefits from motion magnification, which adds an improvement of 3.5 points in accuracy and 7.5 points in F1 score with respect to the combination of both visual and structured features, confirming the contribution of facial micro-movements and their relevance to MMSE estimation.

Besides evaluating our method on MMSE estimation, we extend our analysis to the prediction of scores of individual questions within the MMSE questionnaire. In this setting, the concatenated visual and structured data embeddings are fed to multiple prediction networks, each supervised by the score on a single question. This more granular design allows for the prediction of scores across various cognitive categories assessed by the MMSE, providing a detailed cognitive profile for each patient. Results are reported in Tab. II, and confirm the general trend reported in Tab. I, although the overall accuracy is lower, which can be explained by the increased difficulty of predicting individual question scores.

Finally, we investigate the impact of different temporal resolutions on the quality and utility of the generated MHIs. Temporal granularity plays a central role in how motion information is encapsulated within each frame, affecting the trade-off between information compression and the expressiveness of the motion history. Tab. III shows performance metrics for different temporal lengths of a MHI, in seconds. A small history window results in lower accuracy, as it produces long input MHI sequences which may lead to excessive feature

TABLE I
MMSE PREDICTION ACCURACY USING VARIOUS COMBINATIONS OF TASK DATA (T), EMOTIONS (E) AND VISUAL PRE-PROCESSING TECHNIQUES, INCLUDING FACE STABILIZATION AND CROPPING (FSC), VIDEO MOTION MAGNIFICATION (VMM), AND MOTION HISTORY IMAGES (MHI).

T	E	FSC	VMM	MHI	Sens.	Spec.	Acc.	F1
✓					38.4	88.8	63.6	38.2
	✓				47.0	92.1	69.6	46.3
✓	✓				49.9	94.1	72.0	50.1
		✓	✓	✓	72.8	97.9	85.4	77.2
✓	✓	✓		✓	75.4	98.1	86.8	78.7
✓	✓	✓	✓	✓	82.6	97.9	90.3	86.2

TABLE II
INDIVIDUAL QUESTION PREDICTION ACCURACY USING VARIOUS COMBINATIONS OF TASK DATA (T), EMOTIONS (E) AND VISUAL PRE-PROCESSING TECHNIQUES.

T	E	FSC	VMM	MHI	Sens.	Spec.	Acc.	F1
✓					32.1	83.3	57.7	31.7
	✓				53.8	82.3	68.1	54.0
✓	✓				73.7	84.4	79.1	73.8
		✓	✓	✓	74.8	85.1	80.0	75.0
✓	✓	✓		✓	75.1	85.4	80.3	75.0
✓	✓	✓	✓	✓	75.2	85.5	80.4	75.1

compression due to adaptive pooling; on the other hand, a large window seems to reduce the effectiveness of motion history, since recent motion may overwrite past information. Our choice of a 3-second window is justified by results as a good compromise.

TABLE III
MMSE PREDICTION ACCURACY AT DIFFERENT MHI GRANULARITIES.

MHI(sec)	Sens	Spec	Acc	F1
1	80.0	98.1	89.1	80.9
2	77.5	98.0	87.8	80.9
3 (ours)	82.6	97.9	90.3	86.2
4	80.4	98.0	89.2	83.0
5	70.7	97.5	84.1	73.7

IV. CONCLUSION

Our study demonstrates the potential of a deep learning-based approach for estimating MMSE scores using multimodal data from video recordings of subjects engaged in cognitive tasks. The integration of task performance metrics and emotional states inferred from facial expressions has proven to be effective, with visual features playing a particularly significant role in assessing cognitive capabilities. Our findings underscore the importance of facial micro-movements, as evidenced by the additional accuracy gain through motion magnification techniques. Moreover, the extension of our model to predict individual question scores within the MMSE questionnaire offers a more detailed cognitive profile for each patient, highlighting the versatility and depth of our approach.

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